Problem Set 8: Expectation, Variance and data.table

SPEA V706 Math Camp | Module 8 · Wednesday, August 12

Part A is about moment algebra. **Part B** is about `data.table`.

Part A. Concepts

A1. Moments from a mass function

Let X be the number of doctor visits a person makes in a year.

x	0	1	2	3	4
$\Pr(X = x)$	0.40	0.30	0.20	0.07	0.03

a. Verify that this is a valid probability mass function.
b. Compute $E[X]$.
c. Compute $E[X^2]$.
d. Compute $\text{Var}(X)$ then σ_X.
e. Interpret $E[X]$ and σ_X in a sentence each.

A2. Linearity in action

Keep X from A1. Suppose each visit costs \$120, and a person pays a \$50 annual premium regardless of how many visits they make. Total annual spending is $S = 50 + 120X$.

a. Compute $E[S]$ using E1–E3. Name the rule you use at each step.
b. Compute $\text{Var}(S)$ using V2. Why does the \$50 premium drop out?
c. Compute σ_S. Confirm that $\sigma_S = 120\,\sigma_X$.

A3. Changing units

Daily high temperatures in a city have mean 20°C and standard deviation 5°C. Fahrenheit is $F = 32 + 1.8\,C$.

a. What is the mean daily high in Fahrenheit?

b. What is the variance in Fahrenheit? What is the standard deviation?
c. A colleague says "the standard deviation went up, so Fahrenheit days are more variable than Celsius days." What is wrong with that sentence?

A4. Jensen's inequality with two wages

A worker earns \$10 per hour with probability 1/2 and \$40 per hour with probability 1/2.

a. Compute $E[W]$, then $\log E[W]$.
b. Compute $E[\log W]$.
c. Which is larger? Is that the direction (E5) predicts for a concave function? Explain.
d. In one sentence: if a paper reports that a program raised "average log wages" by 0.10, is that the same as saying it raised average wages by 10%? Say what it is closer to.

A5. A one-line proof

Using only E1 and E3, show that $E[X - E[X]] = 0$ for any random variable X with a finite mean. Say which rule justifies each step, and say in words why the deviation from the mean averages to zero.

Part B. R

Work in a script (`ps2.R`).

```
pacman::p_load(data.table, wooldridge)
data(wage1, package = "wooldridge")
dt <- as.data.table(wage1)
```

B1. Look before you leap

a. How many rows and columns does `dt` have? (`dim()`, `nrow()`, `ncol()`, or `.N` inside the brackets.)
b. Run `str(dt)` and `summary(dt)`. Which variables are 0/1 indicators?
c. Does `wage` have any missing values? (`sum(is.na(dt$wage))`.)

B2. The `i, j, by` model

Recall the shape of a `data.table` call: `dt[i, j, by]`, meaning *filter rows, compute something, within groups.*

a. Using only `i`, count the workers with more than 12 years of schooling.

b. Using only `j`, compute the mean and standard deviation of `wage` in one call that returns a one-row table with named columns.
c. Using `j` and `by`, compute the mean wage separately for women and men.
d. Using all three slots, compute the mean wage by sex **among married workers only**.

B3. Creating columns with `:=`

a. Create a new column `lwage_check` equal to `log(wage)`.
b. The data already has a column `lwage`. Verify your column matches it, for example with `dt[, all.equal(lwage, lwage_check)]`.
c. Create a 0/1 column `hs_plus` equal to 1 when `educ` is at least 12.
d. What did `:=` return, and why is that different from what `mean(wage)` returns in the `j` slot?

B4. Jensen, numerically

You showed in A4 that $E[\log W] < \log E[W]$. Check it on real data.

a. Compute `mean(log(wage))` and `log(mean(wage))` in `dt`.
b. Which is larger? Does it match the direction from A4?
c. Compute the gap between them. Then explain, in one sentence, why reporting `exp(mean(log(wage)))` gives you something smaller than the average wage.

B5. Group summaries and `.SDcols`

a. Compute the mean of `wage`, `educ`, and `exper` all at once, using `lapply(.SD, mean)` with `.SDcols`.
b. Do the same, but grouped by `female`.
c. Compute the mean *and* the standard deviation of `wage` by the combination of `female` and `married`. You should get four rows.
d. Chain a second set of brackets onto (c) to sort the result by mean wage, descending.

B6. Connect it back

a. For the four groups in B5(c), which has the highest mean wage and which the lowest? Report the gap.
b. Each group mean is an estimate of a *conditional* expectation, $E[\text{wage} \mid \text{female}, \text{married}]$. Write that sentence out for the group with the highest mean.
c. You now have four numbers estimated from four different subsample sizes. Which do you trust most, and why?

Check yourself. Without notes:

1. Why is $\mathrm{Var}(X) = E[X^2] - (E[X])^2$ always non-negative?
2. What does adding a constant to X do to its mean? To its variance?
3. In `dt[educ > 12, mean(wage), by = female]`, what does each of the three slots do?